

Execution-order comparison

Pipeline: Normal LLM | 2 runs | 10 identical question texts | Reference: R1

Run	Folder / model
R1	exp1_llm_only_2026-10-09_18-33-19 llama3
R2	exp1_llm_only_2026-10-09_19-05-55 llama3

Main observations

These are observed differences, not a causal estimate of the effect of shuffling. Output length, background activity and measurement variation may contribute.

Recorded model names match; this does not establish identical model versions or generation settings.

Generation settings, token counts and RAG index identity are not consistently recorded by these reports; equality of those conditions cannot be verified.

Lowest observed emissions: R1, 5.1293 g CO₂e. Spread between highest and lowest: 0.0339 g (0.6609% relative to the lowest).

R2 versus R1: different execution sequence. 10 valid successful question pairs; 6 lower, 4 higher, 0 unchanged emissions.

Largest absolute change: Q2, 0.31212 g; execution position 2 to 1. Net change over these matched successful pairs: 0.0339 g.

R2: latency_s change summed over 10 matched successful pairs: 20.714.

Recorded metadata differences: Ordering, Seed, Runtime (s). Unknown or unrecorded values do not establish equivalence.

Historical zero defaults cannot be distinguished from measured zeros. Missing measurements are excluded and partial sums labelled. Percentage changes with a zero reference are N/A.

Run cost and coverage

Observed sums include failed requests. Coverage is valid readings / all questions. Partial sums cannot be treated as complete batch costs.

Run	Emissions (g)	Energy (Wh)	Request time (s)	Failures / unknown
R1	5.1293 [10/10]	N/A (partial) [0/10]	898.39 [10/10]	0
R2	5.1632 [10/10]	N/A (partial) [0/10]	919.1 [10/10]	0

Batch costs: R1-R2

Emissions (g CO2e)

Observed sums across all requests; partial coverage is labelled.

R1		5.129
	Valid 10/10	
R2		5.163
	Valid 10/10	

Energy (Wh)

Observed sums across all requests; partial coverage is labelled.

R1	N/A valid 0/10
R2	N/A valid 0/10

Metadata differences

Different recorded values are marked *. Not recorded means unavailable, not equal.

Setting	R1	R2
Folder *	exp1_llm_only_2026-10-09_18-33-19	exp1_llm_only_2026-10-09_19-05-55
Model(s)	llama3	llama3
Pipeline	Normal LLM	Normal LLM
Ordering *	Keep original order	Freely shuffle all selected questions regardless of question type
Seed *	Not recorded	4186432536
Questions	10	10
Runtime (s) *	898.9362931251526	919.6258676052094
Completion evidence	Summary present	Summary present
codecarbon_version	"3.3.0"	"3.3.0"
os	"Windows-11-10.0.26200-SP0"	"Windows-11-10.0.26200-SP0"
effective_scope	"machine"	"machine"
measure_power_secs	1	1
pue	1.0	1.0
country_iso_code	"IND"	"IND"
config_profile	"inference"	"inference"
configured_settings	{"allow_multiple_runs": true, "country_iso_code": "IND", "log_level": "error", "measure_power_secs": 1, "pue": 1.0, "save_to_api": false, "tracking_mode": "machine"}	{"allow_multiple_runs": true, "country_iso_code": "IND", "log_level": "error", "measure_power_secs": 1, "pue": 1.0, "save_to_api": false, "tracking_mode": "machine"}
hardware	{"cpu_model": "12th Gen Intel(R) Core(TM) i5-1235U", "gpu_count": 0, "gpu_model": "", "ram_total_size": 15.69796371459961}	{"cpu_model": "12th Gen Intel(R) Core(TM) i5-1235U", "gpu_count": 0, "gpu_model": "", "ram_total_size": 15.69796371459961}
backends	{"cpu": "unknown / not exposed by this version", "gpu": "unknown / not exposed by this version", "ram": "unknown / not exposed by this version"}	{"cpu": "unknown / not exposed by this version", "gpu": "unknown / not exposed by this version", "ram": "unknown / not exposed by this version"}
hardware_evidence	[{"class": "RAM", "mode": "unknown"}, {"class": "CPU", "mode": "windows_emi"}]	[{"class": "RAM", "mode": "unknown"}, {"class": "CPU", "mode": "windows_emi"}]

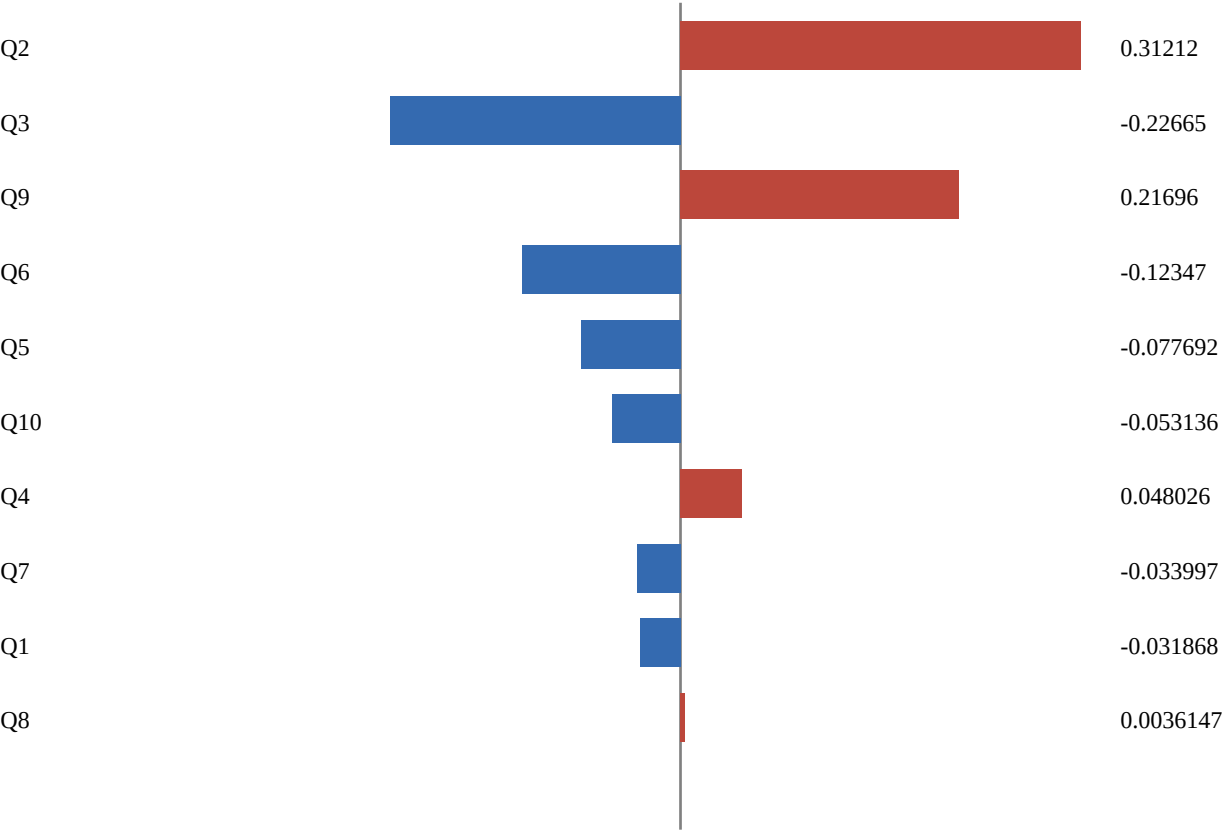
Question emissions change (%)

Relative to R1. Blue = lower, red = higher; grey = missing, failed, or zero reference. Q labels map to exact question text in the comparison table. Color saturates at +/-100%; cell numbers retain the actual change.

	R1	R2
Q1	0	-5.5309
Q2	0	43.497
Q3	0	-34.192
Q4	0	15.813
Q5	0	-15.328
Q6	0	-21.317
Q7	0	-9.2475
Q8	0	0.85673
Q9	0	48.601
Q10	0	-9.7156

Largest changes: R2 versus R1

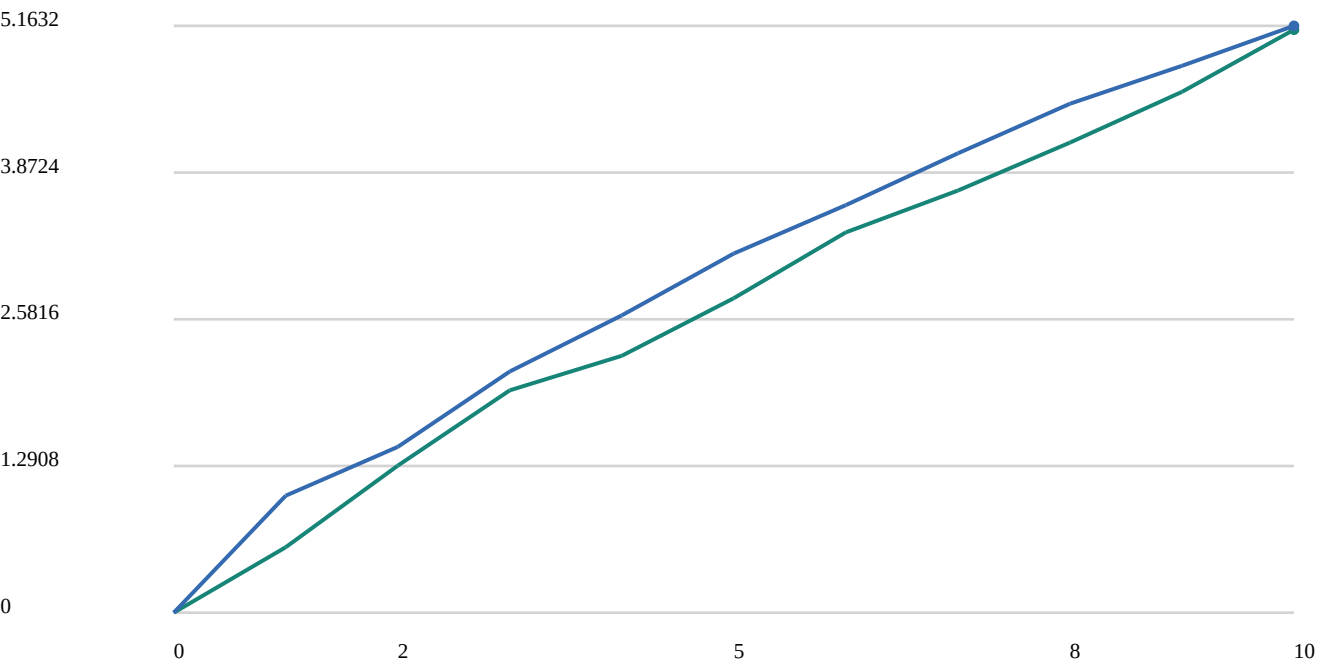
Absolute emissions changes for up to 12 matched successful questions. Left = decrease; right = increase. Labels show g CO2e.



Cumulative emissions through execution

X: number of questions executed. Y: cumulative g CO2e. Question identities at a given position can differ. Curves stop at the first missing measurement rather than implying zero cost. Failed-request measurements are included.

R1: 10/10 questions accumulated
R2: 10/10 questions accumulated



Aligned question comparison

Q identifiers follow the reference run. Text matching is exact. Mean emissions are calculated per question across all selected runs, using recorded values only; missing readings are excluded and the contributor count is shown. Recorded costs for failed attempts are included. Percentage changes are unavailable for a zero reference.

Q1: Outline a step-by-step diagnostic workflow to resolve a multimodal product complaint (e.g., a customer claiming they received a damaged item, with a text description and an image description).

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	1	ok	0.57618	0.56024 (2/2)	N/A	91.469	0	0
R2	5	ok	0.54431	0.56024 (2/2)	N/A	95.995	-0.031868	-5.5309

Q2: Propose a systematic strategy for reviewing a backlog of 500 support tickets to identify a recurring hardware defect, and explain how you would format that data into an actionable product engineering report.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	2	ok	0.71757	0.87363 (2/2)	N/A	122.01	0	0
R2	1	ok	1.0297	0.87363 (2/2)	N/A	170.99	0.31212	43.497

Q3: Formulate a resolution framework for a scenario where two regional shipping policies contradict each other regarding international customs fees. How does a global support agent determine which policy overrides the other?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	3	ok	0.66288	0.54955 (2/2)	N/A	113.12	0	0
R2	8	ok	0.43623	0.54955 (2/2)	N/A	83.156	-0.22665	-34.192

Q4: Imagine a customer writes an angry complaint claiming their newly purchased software license key is showing as 'Invalid.' What is the first diagnostic step?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	4	ok	0.30372	0.32773 (2/2)	N/A	52.533	0	0
R2	10	ok	0.35175	0.32773 (2/2)	N/A	69.66	0.048026	15.813

Q5: If the system shows the key was already activated from a different IP address, how do you communicate this without accusing the customer of fraud?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	5	ok	0.50687	0.46802 (2/2)	N/A	86.786	0	0

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R2	2	ok	0.42918	0.46802 (2/2)	N/A	73.932	-0.077692	-15.328

Q6: The customer responds furiously, demanding an immediate refund. What de-escalation techniques should be used here?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	6	ok	0.57922	0.51749 (2/2)	N/A	101.74	0	0
R2	7	ok	0.45575	0.51749 (2/2)	N/A	84.647	-0.12347	-21.317

Q7: What backend policy checks must be passed before a replacement key or refund can be securely issued?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	7	ok	0.36764	0.35064 (2/2)	N/A	66.208	0	0
R2	9	ok	0.33364	0.35064 (2/2)	N/A	65.351	-0.033997	-9.2475

Q8: How would you categorize and log this specific interaction to improve future automated fraud-detection routing?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	8	ok	0.42192	0.42372 (2/2)	N/A	78.028	0	0
R2	6	ok	0.42553	0.42372 (2/2)	N/A	77.243	0.0036147	0.85673

Q9: Repeated trials of the same operational task can cause agent pass rates to drop — propose a consistency-improvement approach appropriate for a local, resource-constrained LLM agent operating without external APIs.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	9	ok	0.4464	0.55488 (2/2)	N/A	84.044	0	0
R2	3	ok	0.66336	0.55488 (2/2)	N/A	112.63	0.21696	48.601

Q10: In a customer support setting, how should a business weigh the tradeoff between utilizing a heavy multi-agent LLM workflow that yields higher resolution accuracy, versus the reduced operational compute load of a smaller, less accurate standalone model?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	10	ok	0.54692	0.52035 (2/2)	N/A	102.45	0	0
R2	4	ok	0.49378	0.52035 (2/2)	N/A	85.491	-0.053136	-9.7156